

Adapting and Piloting Measurement Tools for Outcomes-Based Financing Programmes for Early Childhood Care and Education

**Learning from Early Childhood Care and
Education Outcomes Funds in Rwanda
and Sierra Leone**

Published by the Education Outcomes Fund

www.educationoutcomesfund.org

comms@edufundmea.org

© 2025. The Education Outcomes Fund

Designed and edited by Green Ink (www.greenink.co.uk)

Contents

Acknowledgements	ii
Abbreviations	iv
Executive summary	1
Five-stage adaptation process	2
Key messages	2
Introduction	4
1. Adapting ECCE measurement tools for OBF programmes	6
1.1 The programmes	7
1.2 Measurement tool selection	7
1.3 Capturing learnings	9
2. The adaptation process	10
2.1 The five stages of measurement tool adaptation	10
2.2 Adapting the adaptation process	19
3. Case study: incorporating disability inclusive adaptations for the ECCE Outcomes Fund in Rwanda	20
3.1 Challenges encountered when adapting for inclusion	20
3.2 Incorporating disability adaptations in the IDELA	20
3.3 Challenges and limitations	21
4. Conclusions	22
4.1 Returning to the objectives	23
4.2 Final reflections	24
Appendices	26
Appendix A. Key resources (non-exhaustive)	26
Appendix B. Glossary	29
Appendix C. Sample adaptations in structural quality tool in Sierra Leone	32
Appendix D. Overview and key lessons	33

Acknowledgements

This technical brief was co-authored by Carolina Laguna from the Education Outcomes Fund (EOF) and Dr. Stuart Cameron, Tanya Kant and Ashna Shah from Oxford Policy Management (OPM).

We acknowledge the contributions of colleagues from OPM who documented lessons learned throughout the measurement tool adaptation process. These include Shafique Arif, Sourovi De, Swapna Nixon, Anjali Pant, Dr. Ankit Saraf, Vipul Singhal and Dr. Laura van der Erve.

We would also like to express our gratitude to Louise Albertyn, Adriana Balducci, Özsel Beleli, Stacie Hartman, Ahrum Kim, Hernan Marisquirena and Jessica Trollip from the EOF team, who generously shared their expertise and reviewed drafts leading up to this publication.

Additionally, we are thankful to our partners for this project whose direct experience piloting the tools shaped the findings of this brief. These include Dalan Development Consultants in Sierra Leone, FATE Consulting Ltd in Rwanda and the team at ECD Measure, whose collaboration was instrumental in our work.

Finally, we appreciate the contributions of our expert reviewers: Dr. Abbie Raikes (founder of ECD Measure and creator of the Brief Early Childhood Quality Inventory), Dr. Sharon Wolf and Ella Wright, who provided invaluable support throughout this process and important feedback to the brief at hand.

General disclaimer

This brief includes contributions from both internal staff and an external consultant. The views expressed are those of the authors and do not necessarily reflect the official policy or position of the Education Outcomes Fund or UNICEF.

This technical brief is part of a series focused on designing outcomes-based financing (OBF) programmes for early childhood care and education (ECCE). This series aims to contribute to the growing knowledge base of OBF for ECCE by sharing insights gained during the design process for ECCE outcomes funds in Rwanda, Sierra Leone and South Africa. This work reflects the EOF's commitment to continuous learning throughout the programme cycle, both to strengthen our own programme design and to grow the collective knowledge base of the OBF and ECCE communities.

The series offers practical guidance for practitioners, policymakers, funders, implementers and technical partners involved in developing and delivering OBF programmes for ECCE.

The series includes:

- ▶ Technical Brief 1: Assessing and Selecting Measurement Tools for Outcomes-Based Financing for Early Childhood Care and Education Programmes
- ▶ Technical Brief 2: Adapting and Piloting Measurement Tools for Outcomes-Based Financing Programmes for Early Childhood Care and Education
- ▶ Technical Brief 3: Designing for Disability Inclusion: Insights from the Early Childhood Care and Education Outcomes Fund in Rwanda
- ▶ Technical Brief 4: Designing Impact Evaluations for Early Childhood Care and Education Programmes using Outcomes-Based Financing
- ▶ Technical Brief 5: Pricing Outcomes in Outcomes-Based Financing for Early Childhood Care and Education

Abbreviations

BEQI	Brief Early Childhood Quality Inventory
ECCE	Early Childhood Care and Education
EOF	Education Outcomes Fund
CAPi	Computer-Assisted Personal Interviewing
IDELA	International Development and Early Learning Assessment
IRR	Inter-Rater Reliability
MINEDUC	Ministry of Education of the Republic of Rwanda
NMS	National Minimum Standards
OBF	Outcomes-Based Financing
OPM	Oxford Policy Management
RBF	Results-Based Financing

Executive summary

This technical brief summarizes the processes and lessons learned from adapting early childhood care and education (ECCE) measurement tools for use in outcomes-based financing (OBF) programmes. ‘Measurement tool adaptation’ refers to the process of preparing data collection instruments such as surveys, structured observations and tests for use in a different context, culture or population to ensure their continued relevance and reliability. This may entail translating the tools into a different language, modifying questions or their wording and adapting their underlying constructs to ensure contextual appropriateness. Crucially, successful measurement tool adaptation involves working with core stakeholders, such as government and programme implementers, to align on what will be evaluated and how.

Contextualized measurement tools are vital to the success of programmes that aim to improve access to quality ECCE. When measurement tools are robustly adapted to the programme’s setting, they enable stakeholders to set specific, measurable objectives and track progress toward achieving them. For example, adapted ECCE measurement tools can assess the baseline quality of centres in a given setting, providing insight into local needs. They can also track child development outcomes over time in a way that is meaningful and feasible within that context. Their role becomes even more critical in OBF programmes, where payments to implementers depend on measurable and verifiable progress toward predefined outcomes.

Ahead of implementing the world’s first outcomes funds¹ exclusively dedicated to ECCE in Rwanda and Sierra Leone, the Education Outcomes Fund (EOF) partnered with Oxford Policy Management (OPM) to prepare measurement tools for use in the programmes’ impact evaluations. The table below presents the domains evaluated in the programmes and the tools OPM adapted to measure them.

Domains and adapted measurement tools

Domain	Description	Tool
Structural quality 	The regulable resources in learning settings, such as the physical environment, educator characteristics and number of educators per child.	Composite checklists based on national minimum standards, combined with global frameworks.
Process quality 	Educator practices and interactions with learners.	Adapted from parts of the Brief Early Childhood Quality Inventory (BEQI), an internationally validated, structured observation checklist.
Child development 	Skills such as cognition, executive function, language, motor, socio-emotional development, as well as early numeracy and literacy.	Adapted version of the International Development and Early Learning Assessment (IDELA), an internationally validated holistic test of children’s early learning and development.

¹ Outcomes funds are a mechanism of outcomes-based financing. To learn more, read *New Information Brief on Outcomes Funds for Education*, EOF (2024).

Five-stage adaptation process

The structural quality, process quality and child development measurement tools were independently adapted to the programme contexts in both Rwanda and Sierra Leone. An adaptation process was developed that would: (i) align the measurement tools with national policies and systems; (ii) maintain the psychometric robustness of globally validated tools; (iii) ensure they were usable in the programme settings; (iv) meet specific demands of OBF such as transparent payments. The resulting process had five stages.

- ▶ **Stage 1: Document review.** The proposed tools (structural quality, process quality and child development measurement tools) were reviewed and adapted in consideration of national policy documents. This was to improve alignment with the ECCE policies prioritized by national education systems in each country. This stage involved initial meetings with programme stakeholders, a review of national standards and curricula, and drawing up an initial list of items that could be added or adapted in the tools.
- ▶ **Stage 2: Stakeholder consultations.** Item-review workshops were conducted with diverse stakeholders to gather detailed feedback on each tool's applicability and relevance to local contexts. This feedback served as the basis for tool adaptations.
- ▶ **Stage 3: Pre-testing.** The tools were tested in a limited number of centres to identify practical challenges and to refine the tools' content. This helped ensure that the tools were as feasible and usable as possible ahead of the pilot.
- ▶ **Stage 4: Piloting of the tools.** Data was collected from a larger sample of centres and rigorous psychometric analyses confirmed the tools' reliability and validity.
- ▶ **Stage 5: Finalization and consensus.** Feedback and piloting data were integrated to optimize the tools' reliability, relevance and effectiveness based on real-world use. The adapted tools were then presented to government stakeholders for official validation and to programme implementers ahead of use in the outcomes funds in Rwanda and Sierra Leone.

Key messages

The following practices helped meet the objectives of the adaptation process, producing measurement tools that were relevant, aligned with ECCE policy priorities in Rwanda and Sierra Leone, and feasible for reliable use in the programmes' contexts and in OBF settings.

- ▶ **Include the tools' creators in the process.** When adapting measurement tools to new settings, engaging with the tools' creators is crucial to understanding the tools' original intent as well as the extent and nature of possible adaptations. In this process, international experts who created the process quality and child development tools, as well as the staff of ministries of education responsible for ECCE quality checklists, were consulted to ensure that the tools' intentions were reflected in the final instruments.
- ▶ **Consult diverse stakeholders with the government in the lead.** Broad stakeholder engagement is key to scaling quality ECCE. In Rwanda and Sierra Leone, implementers, technical experts and national partners supported the adaptation process, while government ownership was ensured through final sign-off on the tools. Differences in perspective are common when working with varied stakeholders but can be mitigated by testing multiple adaptation options during the pilot phase.
- ▶ **Pre-test before the pilot.** Informally testing tools in a small number of settings helps identify major usability and contextual issues early. Once a full pilot has been conducted, making significant adaptations becomes difficult. Taking ethically approved pictures of the ECCE centres can also support tool refinement and enumerator training ahead of the pilot.

- ▶ **Treat the pilot like real field work.** It is important to ensure thorough training of enumerators, including reliability training and the development of a full fieldwork model, as well as the careful management and supervision of enumerator teams. It is also important to collect and manage the data for analysis. All these elements are just as crucial in the pilot as in an actual evaluation. Pilot data helps ensure that adapted tools retain their validity and reliability, despite contextual adaptations.
- ▶ **Build consensus.** Stakeholder consensus is critical, especially in the context of OBF programmes where payments are contingent upon the achievement of measurable outcome. Ahead of implementation it is particularly important to align with programme implementers as their interventions will be evaluated with the tools. It is also helpful to engage both managerial and technical government staff with approval authority throughout the process and to clearly communicate how their input has been incorporated in the final tools. This fosters strong buy-in and reinforces trust in the tools, supporting effective programme implementation and evaluation.

These practices, alongside the five-stage framework, aim to offer actionable and adaptable guidance for others undertaking similar measurement tool adaptations. Sharing the insights generated from this adaptation process reflects EOF's broader commitment to reflective learning and continuous improvement in designing outcomes-based ECCE programmes. As the structural quality, process quality and child development measurement tools are implemented to evaluate and learn from the ECCE outcomes funds in Rwanda and Sierra Leone, EOF will continue to document and share learnings. Through these briefs (and those to come), we seek to inform ongoing dialogue among technical experts designing ECCE and OBF programmes to collectively enhance the measurement of ECCE outcomes and OBF programme delivery.

Introduction

Early childhood care and education (ECCE) is a critical investment in the foundations of lifelong learning and well-being.² Yet the benefits of ECCE remain inaccessible for most children in low- and middle-income countries (LMICs) due to growing financing gaps alongside access and quality barriers.³

Recognizing the urgent need to expand access to equitable and quality ECCE, yet faced with strained education budgets, governments are increasingly exploring innovative financing mechanisms. These include the governments of Rwanda and Sierra Leone, who have partnered with the Education Outcomes Fund (EOF) to establish the world's first outcomes funds exclusively dedicated to ECCE. Outcomes funds are a form of OBF where funding is pooled to make outcomes-linked payments to multiple programme implementers working towards the same, predefined outcomes.⁴

What is OBF?

Whereas traditional ECCE programmes pay for inputs (e.g. infrastructure or learning materials) or activities (e.g. teacher training or nutrition services), OBF links payments to programme implementers to the achievement of predefined outcomes.

Throughout an OBF programme, the predefined outcomes are measured and verified by independent evaluators. Their evaluations trigger payment to implementers if, and when, targets are met. Held accountable for outcomes rather than inputs, implementers can innovate and adapt their interventions in real time, benefiting from data-driven insights generated from frequent outcomes measurement.⁵

In the context of ECCE, this can involve measuring a combination of:

- ▶ **directly observable outputs** such as the number of ECCE centres that open and the number of children they serve
- ▶ **intermediate outcomes** such as the quality of ECCE services
- ▶ **final outcomes** such as improvements in child development.

Rigorous and reliable measurement is essential for setting goals and tracking progress in any ECCE programme. In OBF programmes it is especially critical, as payments depend on the achievement of measurable results. While a growing number of tools are available to measure ECCE quality and child development outcomes, none have been specifically designed for use in OBF.⁶ Among those deemed suitable for OBF impact evaluations, fewer have undergone validation in low- and middle-income countries (LMICs), posing a significant challenge to scaling outcomes-based approaches to ECCE in diverse contexts.

Adaptation is critical to the reliability of any measurement tool. A tool developed in one setting, such as a high-income country, may not be appropriate or effective elsewhere, due to differences in language, educational systems, or child development norms. Therefore, measurement tools must be adapted

² Draper et al. (2024).

³ UNESCO (2024).

⁴ EOF (2024).

⁵ To learn more about the case for OBF in ECCE, refer to the *Outcomes-Based Finance and Early Childhood Care and Education Concept Paper*, EOF (2025a).

⁶ To learn more about ECCE measurement tools, refer to *Measuring Child Development and Early Learning*, Early Learning Partnership (2016).

before use in new cultural, linguistic or contextual settings. This process goes beyond translation; it may involve re-wording items for clarity, adjusting constructs to reflect local practices and ensuring the tool resonates with the lived experiences of the target population, while testing that the tool remains psychometrically sound.

This technical brief contributes to the growing dialogue on OBF in ECCE and aims to support the advancement of effective ECCE programming more broadly by documenting the five-stage adaptation of quality and child development measurement tools led by OPM in Rwanda and Sierra Leone.

1. Adapting ECCE measurement tools for OBF programmes

The adaptation of ECCE quality and child development measurement tools for use in the ECCE outcomes funds in Rwanda and Sierra Leone was guided by four core objectives:

► **Objective 1: aligning measurement tools with national policies and systems**

A central objective of the tool adaptation process was to align the measurement tools with the national policies, curricula and quality standards of Rwanda and Sierra Leone. In the short term, this ensures that implementers are incentivized to meet goals that are relevant to national governments' stakeholders. In the long term, it lays the groundwork for institutionalizing tools within national quality assurance systems. This objective was especially key for EOF in supporting the broader system-strengthening goals of the outcomes funds. When done well, aligning tools with national policies and systems preserves the tools' technical integrity, while enabling their re-use across programmes in the same setting, creating lasting value for national systems.

► **Objective 2: maintaining psychometric soundness when adapting to national contexts**

Measurement tools aim to objectively measure latent constructs — underlying skills, qualities, or behaviours that cannot be observed directly. These constructs can be inferred through other observable tasks or behaviours, which form the items in a psychometric tool. When a tool has been psychometrically validated, it has been shown to measure latent constructs' validity consistently across time, assessors and settings reliability.⁷

Two tools selected for the programmes were psychometrically validated. While this strengthens rigour and comparability, it also constrains the scope for adaptation. Before implementing these tools in a new setting — particularly following translation and contextual adaptation — it was essential to confirm that key criteria, such as alignment with research-based sub-domains, remained intact so that the tools continue to measure what they were designed to assess.

► **Objective 3: ensuring the tools' usability in new settings**

To ensure reliable use by enumerators, it was crucial that the tools reflected the contextual realities in Rwanda and Sierra Leone, such as their physical landscapes, resource constraints and data availability. Enumerators — responsible for systemically collecting data by administering the tools — undergo training ahead of data collection but are not necessarily experts in early childhood development. This underscored the importance of adapting tools with clear, user-friendly formats that could capture meaningful data during programme implementation and evaluation.

► **Objective 4: meeting the specific demands of OBF**

In OBF programmes, regular, rigorous measurement of predefined outcomes (i) triggers payments to implementers when targets are met and (ii) supports continuous learning for implementers and governments throughout and after implementation. This high-stakes measurement requires strict alignment between tools and programme goals as well as advance stakeholder agreement on the final instruments. Because of the financial stakes involved, measurement tools had to be carefully and reliably adapted before use in new OBF settings to minimize the risk of misaligned incentives and support fair and transparent payment structures.⁸

⁷ Please refer to the [Glossary](#) on page 29 for definitions and key terms.

⁸ Holler et al. (2020); Gustafsson-Wright et al. (2016).

1.1 The programmes

The ECCE outcomes funds in Rwanda and Sierra Leone are designed around predefined outcomes aligned with each government's ECCE policy objectives. While both programmes focus on quality and child development outcomes and use a similar set of measurement tools, they operate in distinct contexts with country-specific ECCE targets. For this reason, two parallel adaptation processes were conducted, one in each country, led by a single OPM team. This enabled lessons to be shared across both settings.

ECCE Outcomes Fund in Rwanda

The Rwandan government has made significant strides towards improving early childhood development outcomes in recent years. Net pre-primary enrolment reached 39% in 2023 and there are established early childhood development standards. Despite these advances, only 12% of children under five years meet developmental targets in literacy and numeracy. Rwanda has prioritized ECD as a key pillar of its National Strategy of Transformation. Expanding both formal pre-primary and ECD centres, with alignment of standards and provision of age-appropriate, play-based learning materials, is an objective in its education sector strategy.

The ECCE Outcomes Fund in Rwanda, Nkuza Neza, aims to bridge the gap in quality between formal and informal pre-primary sectors by improving the quality of existing, community-based early childhood centres in rural areas serving three- to five-year-olds. There is a special focus on the inclusion of children with disabilities in these centres. Towards this aim, the programme's implementers will be held accountable (and financially incentivized through OBF) to show progress towards the following outcomes: improvements in the centres' structural and process quality and improvements in holistic child development outcomes, with a particular incentive for the inclusion of children with disabilities.

ECCE Outcomes Fund in Sierra Leone

Sierra Leone has tripled pre-primary education enrolment from 7% in 2011 to 24% in 2023 and has introduced policies to expand access to quality ECCE. This includes the National Policy on Integrated Early Childhood Development, which has made one year of pre-primary education compulsory. Policy goals for the sector include: radical inclusion, expanding access to underserved children; provision of quality ECCE including through safe and accessible spaces and teacher training in both formal and non-formal settings; and development of age-appropriate curriculum and teaching materials, including play-based learning. Despite the recent progress, most children still do not have access to ECCE, and half of children aged three to four are below the expected child development milestones.

The focus of the ECCE Outcomes Fund in Sierra Leone, Salone Pikin 4 Play en Lan (SPPL), is, accordingly, on expanding access to quality ECCE for three- to five-year-olds through the opening of new, community-based centres in remote, low-income areas with no access to pre-primary education. Implementers will be contracted to achieve the following predefined outcomes: meeting minimum safety standards for opening centres; increasing access to the centres as measured through attendance; and improving holistic child development outcomes through improving the centres' structural and process quality.

1.2 Measurement tool selection

Selecting appropriate measurement tools for use in ECCE outcomes funds required balancing multiple priorities. For instance, the tools needed to closely align with the programme goals, meet OBF requirements and align with national standards and government preferences while remaining reliable for use in diverse ECCE settings. The following criteria considered during the measurement tool selection process reflect the unique demands on measurement posed by OBF and the programmes' contexts:

- ▶ Because payments are tied to independently verified outcomes, tools had to be suitable for rigorous impact evaluations, with clear, objective indicators.
- ▶ The financial incentives embedded in OBF can raise the risk of misaligned incentives such as gaming or manipulation, making it essential to select tools that minimize subjectivity and reduce opportunities for bias. For example, tools relying on self-reporting, such as teacher surveys, were not considered for the programmes.
- ▶ Tools needed to be adaptable across formal ECCE centres, home-based care and community-based programmes. Additionally, tools had to be grounded in the realities of low-resource settings to ensure operational relevance and feasibility.
- ▶ Where possible, tools should have been developed and studied using psychometric analysis to establish their reliability and validity across different international contexts. This helps to ensure that results are interpretable, trusted and actionable for government and programme stakeholders.

In collaboration with an advisory panel of ECCE measurement experts, EOF selected a combination of national checklists and internationally validated measurement for use in the ECCE outcomes funds in Rwanda and Sierra Leone. EOF first reviewed the ECCE measurement tools available in each country and internationally recognized tools, shortlisting those suitable for the context and for use in OBF.⁹ Government preferences played a central role in the subsequent selection process to ensure the tools were both contextually appropriate and aligned with national priorities. Table 1 outlines the tools chosen from the shortlist to measure the predefined outcomes in both programmes along with the rationale for each selection.

Table 1. Measurement tools selected

Domain	Structural quality	Process quality	Child development
What is being measured?	The regulable resources in the learning setting, such as educator qualifications, child-to-educator ratios and the physical environment.	The nature of daily interactions and experiences within the learning environment, including the quality of educator-child interactions, emotional support and instructional support.	Rather than a narrow focus on pre-academic outcomes, holistic child development includes measures of cognitive and executive function skills, language and communication skills, socio-emotional development, motor skills and early literacy and numeracy.
How it is being measured?	Composite checklists were developed for each programme, with most items drawn from Rwanda and Sierra Leone's national minimum standards for ECCE. To ensure coverage of generally agreed-upon domains of structural quality, some items were added to the checklists from international tools, including the World Bank's Teach ECE tool and structural elements from the Brief Early Childhood Quality Index (BEQI).	Process-related elements of BEQI, developed by ECD Measure. BEQI is an observation tool with a 50-item yes/no checklist that examines provider-child interactions, activities, materials and health and safety indicators. 39 items are related to process quality.	The International Development and Early Learning Assessment (IDELA) developed by Save the Children. IDELA measures the developmental outcomes through direct assessment. It includes 24 core items with additional optional items, typically requiring 30 minutes per child.

⁹ To learn more about the tool selection process, refer to *Selecting Measurement Tools for OBF Programmes for ECCE*, EOF (2025b).

Domain	Structural quality	Process quality	Child development
Sub-scales	▶ Basic sanitation ▶ Class size/ratio ▶ Safety ▶ Space and materials ▶ Staff characteristics	▶ Play-based learning ▶ Learning through conversation ▶ Promoting strong relationships¹⁰	▶ Emergent literacy ▶ Emergent numeracy ▶ Social-emotional skills ▶ Motor skills ▶ Executive functioning¹¹
Why was this tool selected?	The measurement advisory panel convened by EOF considered international evidence and theory about what constitutes structural quality and used this to identify gaps in national standards. As certain sub-domains were under-represented in the government tools, the checklist was supplemented with elements of BEQI and Teach ECE to ensure the comprehensive and balanced assessment of structural quality.	BEQI was selected because it is simple to train on, easy to use and produces straightforward data. ECD Measure encourages users to review and adjust items to align with local priorities and relevant standards. This flexibility allows for context-specific customization and contextual modifications.	IDELA was selected as it balances cross-country applicability, feasibility of administration in low resource settings and psychometric rigour. IDELA has been validated in multiple LMICs, including Rwanda. ¹² It is an open-source instrument that is easily adapted to different contexts and includes adaptation and administration guidance for each question.

1.3 Capturing learnings

OPM systematically documented take-aways during the five stages creating summative reports and structured reflections to help support the efforts of policymakers, practitioners and researchers engaged in their own tool adaptation processes. Twelve trained enumerators (six per country) provided feedback through pilot fieldwork reports and regular debriefings. Their input highlighted practical challenges, contextual nuances and suggestions for improving the tools and their implementation. Additionally, eight adaptation team members (six from OPM and two from EOF) completed a reflective survey to capture the insights, challenges and outcomes from each adaptation stage. Their generous contributions serve as the basis for the reflections in this brief.

The rest of the brief is structured as follows:

- ▶ **section 2** outlines the five-stage adaptation process, with objectives, examples and best practices shared for each stage
- ▶ **section 3** provides a case study on adapting measurement tools for disability inclusion in Rwanda
- ▶ **section 4** shares conclusions and reflections to support future ECCE and OBF tool adaptation processes.

¹⁰ To learn more about the theoretical grounding of these domains, refer to [Appendix A. Key resources](#).

¹¹ Ibid.

¹² Ibid.

2. The adaptation process

2.1 The five stages of measurement tool adaptation

Measurement tool adaptation requires meticulous planning, robust stakeholder engagement and iterative testing and refinement to ensure that tools are both reliable and contextually relevant. This is especially critical in OBF settings, where tools directly influence funding decisions and must meet high standards of reliability to ensure fairness. To meet these demands, OPM led a five-stage adaptation process for ECCE centre quality and child development measurement tools in Rwanda and Sierra Leone. The following section outlines each stage, its specific goals, illustrative examples and emerging best practices.

Table 2. The five adaptation stages

	Stage 1: document review	▶ 1 month
	Stage 2: stakeholder consultations	▶ 1.5 months
	Stage 3: pre-testing	▶ 2 weeks
	Stage 4: tool piloting	▶ 2.5 months
	Stage 5: final refinements and consensus	▶ 2 months



Stage 1: document review

This stage focused on ensuring the tools' coherence with each country's ECCE standards, objectives and curricula in order to meet a primary objective of the adaptation process; that is, aligning measurement tools with national policies and systems. More specifically, OPM reviewed key documents such as national minimum standards and ECCE curricula and held initial meetings with programme stakeholders. This allowed the adaptation team to identify where the selected tools aligned with national systems and where adaptations or additions were needed. This stage not only ensured that selected tools captured national education system priorities but increased the tools' relevance and built stakeholder confidence in the tools by demonstrating their value in supporting national ECCE goals.¹³

Two types of reference documents were reviewed to help meet the objectives of this stage:

- ▶ **National Minimum Standards (NMS):** the Rwanda National Standards and Guidelines for ECD¹⁴ and the Sierra Leone National Minimum Standards for ECCE¹⁵ were consulted to understand the broader expectations of government around ECCE quality in the two countries. These documents had already been consulted by the measurement experts who prepared the original structural quality checklists.
- ▶ **ECCE curricula** were used to adapt the process quality and child development tools. In Rwanda, this was a pre-primary curriculum mainly focused on formal pre-primary education. In Sierra Leone, it was the curriculum framework corresponding to the NMS.

¹³ Winthrop and McGivney (2016).

¹⁴ Republic of Rwanda, National Child Development Agency (2024).

¹⁵ Government of Sierra Leone, Ministry of Basic & Senior Secondary Education (2020).

Inclusion of additional items based on a national document review

A review of national policy documents prompted the expansion of the original tools to better align with country-specific standards and priorities. OPM developed a set of potential new items for discussion and piloting in subsequent stages of the adaptation process. These additions were informed by gaps identified during the document review.



Structural quality tools: composite structural quality tools were developed by measurement experts using items drawn from NMS, supplemented with elements from international tools to ensure balance across important structural quality domains. This stage of the adaptation process involved a second review of national policy documents, focusing on closer alignment with national priorities and identifying areas for stakeholder consultation.

Based on this second review, some additional items were developed for the structural quality tool and others were re-phrased to enhance their suitability for an observation-based tool. For example, in Sierra Leone, the NMS item: “Is the ceiling properly done without leakages?” was added but re-phrased to: “Are there leaks or holes in the roof/ceiling?” to support objective data collection.



Process quality tool: the original process quality tool was based on the domains and items from the BEQI tool. The national standards documents were reviewed to identify process-related elements not captured by BEQI. For instance, Rwanda’s standards emphasized play-based learning and hygiene, and Sierra Leone’s standards highlighted active learning and inclusive interactions. These standards were generally consistent with BEQI’s focus on play-based learning, learning through conversations and strong relationships, both among learners and between learners and educators.

In Sierra Leone, additional concepts from the NMS such as promoting healthy habits and providing opportunities for the appreciation of local culture and values were considered for inclusion. ECD Measure identified potential additional items to cover these areas, many of which had been piloted in other contexts as part of BEQI but had not been part of the standard checklist. These items were earmarked for further discussion and testing in the subsequent stages of the adaptation.



Child development tool: the IDELA assessment served as the starting point for the child development tool. A review of the ECCE curricula in Rwanda and Sierra Leone was conducted to assess IDELA’s alignment and to identify areas of enhancement. In Sierra Leone, the ECCE curriculum emphasized some sub-domains not fully covered by IDELA such as: religion (knowledge of prayers, religious routines); rhyming (use of songs and poems as a way of teaching); and health and self-care. In Rwanda, the curriculum highlighted the use of basic spoken English in early learning activities. These areas were flagged for stakeholder discussion in Stage 2 as potential domains for the development of additional items.



Best practice for aligning measurement tools with national systems: consult with the creators of the tools during adaptation

It is helpful to engage with the creators of a tool from the early stages of adaptation to understand how it can be modified and aligned with national objectives without compromising its validity. It is also useful to get their feedback on any adaptations based on their experiences in similar contexts. In Rwanda and Sierra Leone, OPM worked with ECD Measure, the developer of the BEQI tool, and IDELA experts who helped ensure that the proposed adaptations were measurable and consistent with the tools’ psychometric frameworks. The team also referred to Save the Children’s IDELA’s Adaptation and Administration Guide, which informed decisions at all stages.¹⁶

¹⁶ International Development and Early Learning Assessment, Save the Children (2019).

Stage 2: stakeholder consultations

Government-led stakeholder consultations are a formal and essential part of the tool adaptation process, ensuring both contextual relevance and local ownership. While consultations and discussions with different stakeholders happened throughout the adaptation process, both virtually and in-person, the most important consultations took place during this stage through a set of dedicated item review workshops in each country. After integrating elements from national standards and ECCE curricula and identifying potential gaps in the original tools, these workshops, led by the respective Ministries of Education, allowed for detailed feedback on the tools' individual items. This reinforced a participatory approach to tool adaptation that was crucial for developing valid, widely accepted measurement tools for use in OBF.¹⁷

Item review workshops fostered alignment and relevance

In each country, a three-day workshop was structured to review one tool per day, using a participatory approach to gather contextual feedback and build local ownership. Participants were given time to review printed tools, followed by structured discussions focused on item relevance and feasibility in ECCE settings. In Sierra Leone, 33 participants were selected by the Sierra Leone Ministry of Basic and Senior Secondary Education (MBSSE); these included government officials, curriculum experts and practitioners. In Rwanda, the Ministry of Education (MINEDUC) brought together 16 officials from various departments.

These sessions fostered open dialogue, enabling real-time revisions where consensus was reached and facilitating negotiations where disagreements arose. Beyond technical input, the process helped ensure alignment with national priorities and encouraged long-term uptake. Most discussions centred on clarifying items and incorporating cultural context.

Adapting the child development tool to the Rwandan context

In Rwanda, stakeholder consultations during this stage revealed priority adaptations to the child development tool, especially given the programme's focus on including children with disabilities. One day of the tool adaptation workshop focused on refining the IDELA tool to reflect the local language, as well as learning priorities and inclusion goals. Participants reviewed existing Kinyarwanda versions of IDELA items and compared multiple translations, where available, to select the most appropriate. Two key adaptations emerged:

- ▶ **Measuring emerging English skills:** the document review had identified the importance of learning some basic knowledge of English in the Rwandan ECCE curriculum. This was discussed with stakeholders, who agreed and recommended adding items to capture children's emergent English skills, as exposure to English is common in Rwandan ECCE centres. Based on the consultations, four items were added to assess basic English comprehension and expression, aligning with national goals around school readiness, as primary schools regularly use English as the language of instruction.
- ▶ **Ensuring inclusion:** given the programme's priority on disability inclusion, discussions centred on how to make IDELA more inclusive. While the IDELA Adaptation and Administration guide developed by Save the Children provided basic guidance on how inclusion-related challenges could be addressed, additional suggestions for adaptations were prepared based on gaps identified through an additional skill mapping.

These adaptations, which were later trialled through pilot testing, underscore the value of participatory approaches to tool adaptation, especially for meeting the objective of alignment with national policies.

¹⁷ Kellaghan and Greaney (2020).



Best practices for stakeholder consultations

Who to engage in consultations:

- ▶ Engage a diverse range of stakeholders, including government officials, academics, educators and curriculum developers to get a well-rounded review of the tools.
- ▶ Ensure that government stakeholders could sign off on the final instruments.
- ▶ Prioritize reviews by educators or practitioners who are directly engaged with children for tool translations to ensure practicality and contextual appropriateness.
- ▶ In OBF settings, involve implementers in stakeholder consultations, as they bring valuable contextual knowledge and their involvement can help reduce disagreements at a later stage. However, their involvement must be carefully timed and structured to avoid conflicts of interest and the risk of “teaching to the test”. In Sierra Leone and Rwanda, implementers were consulted after the pilot, rather than as part of the initial government-led workshop.

How to ensure consultation success:

- ▶ Disagreement among stakeholders is common in collaborative settings. To manage these conversations, emphasize that various options can be pre-tested (which is described in Stage 3), reassuring stakeholders that the selection process is thorough and data-driven.
- ▶ Send invitations directly from the Ministry of Education or senior government office to leverage influence and ensure attendance. Official endorsement underscores the event’s importance and alignment with national priorities, encouraging participation.

Stage 3: pre-testing

The purpose of this stage was to bridge the theoretical design of the tools with their practical application in the real-world settings of ECCE centres. “Pre-test” and “pilot” are often used interchangeably. In this brief, pre-testing refers to the initial test of the tools in a limited number of centres, whereas the pilot represents a full test of the tools in a larger number of centres during which data is gathered for analysis. Comprehensive feedback regarding the daily operational schedule of the centres was also collected during this stage, allowing for logistical adjustments that supported the objective of ensuring the tools’ practical usability.

Pre-testing provides a valuable addition to adaptation processes, as there is often limited scope for adapting the tools post-pilot due to budgetary and time constraints. Where possible, pre-test centres should be selected to reflect a range of quality levels. This allows the tool to be tested in both higher- and lower-quality settings, helping assess its applicability and sensitivity across diverse contexts. In this project, five centres were selected for pre-testing in each country.

Tool translations

In both Rwanda and Sierra Leone, the structural and process quality tools were administered in English, as enumerators were proficient in the language. Since these tools are largely observational and involve minimal interaction with educators or children, using English reduced the need for translation and helped maintain the precision of instructions and constructs.

In contrast, the child development tool required translation into local languages due to its interactive nature and its focus on children’s language and comprehension skills. Translation ensured accessibility and a more accurate assessment of children’s verbal responses. In Rwanda, IDELA was administered in Kinyarwanda, building on a previously validated version that was adapted to meet the project’s needs

and stakeholder priorities. In Sierra Leone, the tool was newly translated into Krio, Temne and Mende, the country's three largest linguistic groups.

Refinements during pre-testing helped meet usability objectives

An essential aspect of pre-testing was refining the tools' content to ensure clarity and relevance. This involved testing the understandability of the questions for both the enumerators and the respondents, i.e. the educators and the children. Feedback obtained during pre-testing led to significant revisions in the wording of questions, the layout of the tools and the instructions for enumerators, which were critical for avoiding confusion and obtaining accurate data:

- ▶ Some items were re-worded for clarity after observing that the original phrasing caused confusion among enumerators. For example, detailed instructions were added to increase usability of the structural quality tool in Rwanda (refer to [Appendix C](#) for more examples from Sierra Leone).

Original item	Instructions added
How many usable toilets or latrines for children are there?	Count number of holes / seats / stances. Only count toilets/latrines that are both (1) available to students (doors are unlocked or a key is available at all times), (2) functional (the toilet is not broken, the toilet hole is not blocked and water is available for flush/pour-flush toilets).

- ▶ In some cases, feedback from this stage resulted in altering the tools' layout to make it more intuitive and user-friendly for enumerators. For example, the structural quality tool in both countries was reorganized to follow a logical sequence for enumerators, beginning with observations of the outdoor environment, followed by toilets and sanitation facilities and concluding with the classroom setting.
- ▶ The pre-pilot revealed logistical implications to tool administration. For example, the process uncovered details on the timing of lunch breaks, allowing OPM to identify the most suitable periods during the day when a continuous, uninterrupted 90-minute window could be secured for observing process quality effectively.
- ▶ The standard guidelines from IDELA advise that administration time is typically around 35 minutes per child. Pre-tests suggested that more time might be needed in these contexts, notwithstanding the fact that with more enumerator practice and careful fieldwork management, the total time needed to administer this tool will reduce.



Best practice for pre-testing: engage officials, trainers and a diverse range of centres

- ▶ Select a diverse array of ECCE settings to gain insights into tool performance across different contexts.
- ▶ Involve ministry officials in pre-test field visits to facilitate access and expedite permissions.
- ▶ Include the in-country trainer responsible for pilot training in pre-tests to provide insights and to improve training efficiency.
- ▶ With permission, take pictures of centres to help enumerators to practice scoring different items during training.
- ▶ Finalize survey flow and conduct comprehensive rehearsals to address operational questions and prevent confusion during data collection.
- ▶ Maintain rigorous, item-level documentation of the pre-test through daily debriefs and feedback tracking to facilitate effective tool refinement.

Stage 4: piloting

Piloting plays a pivotal role in ensuring tools' practical applicability and reliability before their full-scale deployment in programmes' impact evaluations by collecting robust data on the tool's performance. In both Rwanda and Sierra Leone, the measurement tools were piloted in 30 centres, with around 120 children in each country.

Effective enumerator training underpinned successful pilot testing

During the pilot, training and data collection processes were designed to strengthen reliability. A week-long immersive training for enumerators included role-playing, mock exercises, a field practice day and group discussions to build a shared understanding. Observation-based tools, such as the process quality tool, required expert guidance and reliability training to ensure that the enumerators were trained adequately and passed tests that would make them reliable observers in various qualities of classroom. The adaptation team also provided supervision and conducted regular debriefs during fieldwork to address concerns and challenges in real time.

Despite these efforts, some challenges remained. Items requiring objective responses, such as: "Is there a chalkboard/whiteboard?" were straightforward to code. However, items with wording open to subjective interpretation, such as: "Is the space large enough?" posed difficulties in reaching a common agreement. To address this, the training sessions included targeted practice using real-life examples in order to standardize the group's understanding of the item as much as possible. Any lingering concerns were addressed during a practice field visit.

The pilots also highlighted the need to balance comprehensive measurement with manageable tool design. Limiting the number of tool items, prioritizing essential data and using user-friendly formats such as clear examples and simplified language were critical to ensure effective implementation in low-resource settings by non-expert enumerators.

Piloting revealed logistical and contextual challenges, helping improve preparedness for full implementation

As in any school survey, the success of the pilot depended on experienced data collection teams who could both overcome and document logistical and attendance challenges encountered during fieldwork. For example, in Rwanda, enumerators documented the additional time that might be needed to conduct IDELA in children's homes, which were quite dispersed and difficult to reach, particularly in mountainous areas. Additionally, the information used for sampling centres in both countries was limited and of variable quality, which required a flexible approach to identify suitable centres. To address these issues, the team streamlined formats and adapted protocols to better align with local routines and environments. These adjustments will help ensure smoother data collection during full-scale implementation, while preserving the integrity of the tools.



Best practices for pilots

The success of the pilot relied heavily on the enumerators' preparedness and training. Key lessons included:

Invest in effective training and practice:

- ▶ Daily practice on all of the tools' items helps refine the tools and the skills of the enumerators, as does employing two full assessments, one using translated tools and one using Computer-Assisted Personal Interviewing (CAPI).
- ▶ Reliability tests and scoring practices, incorporated throughout the pilot, help ensure enumerators were well-prepared.
- ▶ The pre-test and a 'practice' day at the start of the pilot helped identify and resolve practical issues early on, improving the process for the main data collection.
- ▶ More enumerators were trained than would be needed for the fieldwork. This approach allowed reliability and performance-based selection of enumerators and ensured a reserve of trained enumerators in the case any one of the selected team would be unavailable.

Ensure real-time monitoring and adaptation:

- ▶ Field supervisors providing immediate feedback to enumerators allows for quick problem-solving and adjustments during data collection.
- ▶ Having enumerators work in pairs during fieldwork allows for cross-checking data, thus minimizing discrepancies or biases and improving reliability.

Stage 5: final refinements and consensus-building

The objective of this stage was to finalize the measurement tools by integrating feedback from the piloting phase and to ensure stakeholder alignment ahead of implementation. This ensured the tools were accurate, effective and aligned with programme goals. Stakeholder consensus is particularly critical in the context of OBF given the financial implications of measurement.

Statistical analyses helped ensure the adapted tools remained consistent and reliable

At the core of a tool refinement process lie psychometric analyses which help identify any anomalies that could impact the tool's operational validity or reliability. In this process, analyses drew on piloting data to measure the following:

- ▶ **Inter-Rater Reliability (IRR):** this analysis was essential to ascertain the consistency of results provided by different enumerators under identical conditions. Across both countries, the tools achieved strong IRR scores (90%); IRR data collected during the pilot helped refine the tools. If certain items consistently showed low inter-rater reliability, i.e. high inter-rater variability between enumerators, these items could be re-worded or clarified to reduce ambiguity. For instance, the structural quality tool generally demonstrated a high level of agreement between the enumerators in terms of recording observations; however, some items that involved subjective estimation (e.g. suitability of an area for children to walk and play) had lower agreement between the enumerators and were flagged for additional training in future rounds.
- ▶ **Internal consistency and reliability:** the internal consistency estimate (measured using Cronbach's alpha) for the adapted IDELA was found to be 0.81 for Sierra Leone and 0.83 for Rwanda.

Both estimates fall within the “good” range of results and indicate that the tool is functioning well as a reliable measure of specific early childhood development domains in the respective contexts.

- **Floor and ceiling effects:** the pilot did not reveal any major concerns around the presence of floor or ceiling effects in the adapted tools. However, in certain cases where floor or ceiling effects were identified, the decision to drop or retain the item was taken on a case-by-case basis, based on the relevance of the item for the larger construct, specific stakeholder feedback, or future relevance of the item in a scaled-up administration. For example, one of the BEQI core items in Sierra Leone demonstrated a floor effect (i.e. no centres met the item). However, the item was retained due to its underlying contribution to the construct of language-rich environments. In another instance in Rwanda, items in the newly introduced Spoken English domain in IDELA showed the presence of floor effect, yet the decision was taken to retain two items, given the importance of this domain to the stakeholders.

The pilot phase had limitations. Pilot sample sizes were too small and not diverse enough to support more advanced psychometric analyses, such as testing for model fit of the four domain IDELA model.¹⁸ This constrained the ability to fully assess how adaptations may have affected measurement properties. Nevertheless, presentations to government stakeholders and implementing partners at the end of the pilot built confidence in the adaptation process and, therefore, consensus was reached in the tool's ability to measure the target outcomes.

Tools were refined to ensure usability

Areas where enumerators consistently faced challenges were flagged for greater emphasis in future training to improve preparedness. For instance, in a few cases in Rwanda, process quality observations were interrupted by the centre's operating hours, which were interspersed with meal or porridge breaks. Documenting this helped improve preparedness for full-scale implementation by helping build the understanding of the different types of routines present in ECCE centres. This contextual challenge during the pilot experience stressed the need to account for outlier cases while planning for future fieldwork logistics.

Tools were refined to meet the needs of OBF

Certain measures helped ensure the adapted tools remained appropriate to the nature of measurement under OBF without compromising their validity. For instance, certain items added to the tools were designated as non-scoring items, meaning they would not be used to evaluate implementers' progress toward predefined outcomes but rather to gather contextual information for learning purposes. Non-scoring items can generate relevant evidence during and after programme implementation, supporting implementers in developing data-driven, iterative ECCE interventions, as well as learnings that align with national ECCE priorities. Including these items for learning purposes only helped ensure the tools' continued validity while aligning them to national ECCE policy priorities.

In order to mitigate potential misaligned incentives associated with high-stakes measurement such as “teaching to the test,” the child development outcomes tool was reviewed to identify questions suitable for alternative versions within a small item bank. Six such items were selected; for example, a question asking children to identify a letter can be replaced with other letters that appear with similar frequency in the language of test administration. This approach helps preserve the integrity of assessments while supporting more robust and flexible evaluation design.

Final consultations helped meet all objectives

Finally, to secure stakeholder buy-in and ensure alignment with national priorities, the refined tools were presented to government and programme stakeholders in Rwanda and Sierra Leone. In-person, formal government validation workshops confirmed that the tools were consistent with national

¹⁸ ‘Model fit’ refers to how well data from actual subjects (in this case, ECCE centres) conforms to researchers' models, based on theory and prior research. Model fit can be tested using methods such as Confirmatory Factor Analysis, which attempts to describe the variability in a set of items in terms of a smaller number of underlying factors based on the model.

educational policies and were suitable for evaluating ECCE outcomes. In parallel, virtual workshops with the programmes' implementing partners (where the full structural quality checklist and summaries of the process quality and child development tools were shared) enabled implementers to raise final queries and contribute practical insights from the field. This feedback was essential for final refinements and helped build trust and transparency among all stakeholders ahead of the programme's launch.

Table 3 summarizes the tools presented to programme stakeholders and [Appendix C](#) provides examples of adaptations made to the structural quality tool in Sierra Leone.

Table 3. Summary of the adaptations

Outcome	Tool	Adaptations
Structural quality	The finalized checklist comprised 55 items in the case of Rwanda and 54 items in the case of Sierra Leone, spanning four sub-scales: basic sanitation, safety, staff characteristics, and space and materials. The majority of these items were derived from the country's respective standards checklists for early childhood centres.	In both countries: ▶ Indicators and questions were re-worded to improve clarity, making them easier for enumerators to administer. ▶ To further support enumerators, detailed instructions were added. Some questions were split into two to facilitate better observation during data collection, while others were removed due to duplication or difficulty.
Process quality	In both countries, the adapted tool consisted of 41 yes/no checklist items across three domains of measurement: play-based learning, learning through conversations and promoting strong relationships. Eight out of 41 items were retained as non-scoring items (items that would not contribute to the centre's overall score or be used for additional context and learning purposes). Some of these were in the core BEQI tool, while others were added in the adaptation process.	▶ Context-specific language was incorporated and examples were expanded to include familiar materials, such as adding sticks for maths and slates for writing in Sierra Leone. ▶ In Rwanda, wording was streamlined by simplifying multiple verbs in the same question and the term "teacher/educator" was replaced with "caregiver" to better align with national terminology. ▶ Items with potential cultural sensitivities were removed where necessary. For example, in Sierra Leone, stakeholders suggested removing the item: "Are there any religious activities conducted in the classroom?" in the process quality tool due to cultural sensitivities.
Child development	The adapted tools consisted of 25 items in the case of Sierra Leone and 26 items in the case of Rwanda. These covered the following sub-scales: emergent literacy, emergent numeracy, social-emotional skills, motor skills and executive functioning.	▶ In both countries, materials and item presentation were adjusted to better reflect local contexts; for example, replacing a picture of a Caucasian girl with an African boy in both countries and swapping mentions of "apples" for "bananas" in Rwanda, which are more contextually relevant to children. ▶ The tools were translated into local languages; Krio, Temne and Mende in the case of Sierra Leone and Kinyarwanda in the case of Rwanda. ▶ In Rwanda, two items were added to assess basic English skills taught in ECCE centres; disability inclusive adaptations were also implemented. ▶ In Sierra Leone, recommendations were made to use chalk and slates instead of paper and pens to better match classroom resources.



Best practices for consensus-building

- ▶ Ensure the right government stakeholders are invited to the final consultation. The best mixture of managerial and technical staff may vary between contexts but it needs to include stakeholders in a position to sign off on the final tools.
- ▶ Keep track of how stakeholders' views expressed in the earlier consultation were included in the final design.
- ▶ Manage expectations about additional changes; ideally, very few changes should be made at this stage because it will no longer be possible to pilot them.
- ▶ Communicate carefully and clearly about the rationale for the instruments and the ways they have been adapted.
- ▶ Listen to concerns and queries about the instruments and be ready to answer these, but also have a back-up plan for the possibility that stakeholders reject the instruments as they are.
- ▶ Consider how the instruments align with national policies and frameworks, such as minimum standards and curricula, with a clear rationale for any departures from these. Some planning may be needed about communication of these points.

2.2 Adapting the adaptation process

The five-stage adaptation process used in the ECCE outcomes funds offers a replicable and flexible model for adapting measurement tools, both for OBF impact evaluations and for broader programme evaluations. While the process was resource-intensive, it proved essential for ensuring that tools were technically robust, contextually relevant and aligned with national systems. In future applications, especially where resources are constrained, there may be opportunities to streamline certain stages. The following “shortcuts” could be considered:

- ▶ **Compress the time for consultations with stakeholders;** this is possible provided that key respondents are able to put aside time to provide rapid feedback.
- ▶ **Reduce the analysis of pilot data;** a more lightweight or rapid adaptation process might opt to prioritize analyses that are most feasible and informative given the sample size (such as IRR, internal consistency and item-level response patterns), while intentionally skipping analyses that require larger and more representative samples to yield reliable conclusions. This approach ensures that the pilot remains focused and practical, while still generating actionable insights for tool refinement.
- ▶ **Roll the pilot into the main evaluation;** a short pilot could be conducted as part of the first phase of the evaluation itself, allowing for minor changes to tools. However, this is high-risk, as it might be too late to make major changes should any major problems arise.

While these shortcuts may offer efficiencies in certain contexts, they come with trade-offs with regards to rigour, reliability and stakeholder engagement. The experience of adapting tools for the ECCE outcomes funds in Rwanda and Sierra Leone suggests that investing in a structured, participatory adaptation process, especially for tools tied to financial incentives, is worth the time and resources. Ultimately, the decision to streamline should be guided by the programme's goals, the complexity of the tools and the importance of stakeholder trust in the measurement process.

3. Case study: incorporating disability inclusive adaptations for the ECCE Outcomes Fund in Rwanda

3.1 Challenges encountered when adapting for inclusion

In Rwanda, a key government priority for the programme and adaptation process was to ensure inclusion of children with different functional abilities. As part of this commitment, children in the programme will initially be screened using the Washington Group/United Nations Children's Fund Child Functioning Module (CFM) to identify children with disabilities and those who will need adaptations for the IDELA assessments. While this screening will generate valuable data, it does not offer guidance on how assessments should be adapted for children with specific impairments or communication differences. In turn, the three measurement tools selected for the programme were reviewed in consultation with ECCE disability inclusion experts:



Structural quality: Rwanda's national frameworks already include criteria such as “disability-friendly” environments, accessible toilets and provisions for children with developmental delays or disabilities. However, these criteria often require further clarification to ensure reliable observation by non-specialist enumerators. Indicators in the tool were simplified, e.g. accessibility focused on the absence of steps in classrooms and play areas and adapted learning materials were referenced generally for sensory impairments without specifying format (e.g. Braille or audio materials). Importantly, government stakeholders emphasized that inclusive infrastructure and materials should be in place regardless of whether children with disabilities are currently enrolled, ensuring readiness for future inclusion.



Process quality: strong ECCE practice should meet the needs of children with diverse physical and cognitive abilities, especially in mixed-age settings where there are multiplied demands on the educators' time and attention.¹⁹ The BEQI tool incorporates this principle. Additional items were piloted to assess whether educators treat all children equitably; however, these did not yield additional insights beyond what was already captured by existing items. This points to a need for future process quality measures to move beyond observable equality to capture children's experiences of belonging, participation and empowerment within ECCE settings.



Child development: the IDELA Adaptation and Administration guide²⁰ includes basic guidance for adapting items for certain disabilities and encourages the inclusion of children with developmental delays. However, this guidance is not all-encompassing. In particular, IDELA's reliance on oral administration poses barriers not only for children with severe hearing impairments but also for neurodiverse children and those using non-oral communication methods. This oral-verbal bias structurally excludes a wide range of children, underscoring the need for alternative assessment tracks, communication-affirming tools, or a broader reconceptualization of “school readiness” in inclusive ECCE models.

3.2 Incorporating disability adaptations in the IDELA

To address the perceived gaps in IDELA, a comprehensive set of disability adaptations were developed for the sub-scales where additional adaptations were possible. These were based on the existing

¹⁹ OECD (2015).

²⁰ International Development and Early Learning Assessment, Save the Children (2019).

guidance provided in the IDELA adaptation manual and were incorporated into the tool during the pilot. Ahead of this, a mapping exercise was conducted to link specific IDELA sub-items to the types of adaptations needed for various impairments or communication differences. Informed by the IDELA Adaptation and Administration Guide and pilot, disability adaptations were developed for most IDELA items and included modified assessment materials, offering alternative response methods (e.g. gestures) and allowing flexibility in how children demonstrated competencies. Additional observational questions were included alongside caregiver responses to the CFM to better identify functional difficulties that could affect IDELA assessment participation.

During the programme's impact evaluation, enumerators will administer six screening questions prior to the IDELA. These will allow them to determine whether adaptations are needed or whether certain items should be respectfully skipped based on functional barriers that cannot not be adequately accommodated. Key features of the IDELA disability adaptation include:

- ▶ **Adaptations developed for most IDELA sub-tests;** the disability adaptations covered a broad range of IDELA items, particularly for mild and severe visual difficulties and motor impairments.
- ▶ **Ease of adoption;** the suggested disability adaptations were kept simple, making use of easily available materials (e.g. paper and cardboard) to ensure that adaptations could be easily incorporated by a wide range of enumerators without requiring specialised equipment. Additional training ensured that the adaptations could easily be administered, even by enumerators who did not have any prior specialization in disability work.

3.3 Challenges and limitations

- ▶ **Digital administration;** the use of a digital platform (CAPI) helped standardize administration of adaptations and automate item-skipping based on enumerator inputs. However, the pilot was conducted entirely in a digital environment and additional challenges could arise if transitioning to an offline data collection model.
- ▶ **Small sample;** during the pilot, the disability adaptations were applied to only two children with mild visual difficulties. This very limited sample did not allow for any detailed analysis of the developed adaptations. Further piloting and iterative refinement are essential to ensure that adaptations are effective across a broader diversity of disabilities and communication styles.
- ▶ **Coverage limitations;** while the adaptations addressed visual and motor impairments, no viable adaptations could be developed for severe hearing impairments within the current IDELA structure due to its dependence on oral instructions. Furthermore, the adaptations did not sufficiently address communication diversity among neurodiverse children or children with more complex developmental profiles. Future adaptations will need to expand beyond functional classifications to incorporate diverse, rights-based understandings of inclusion.

Disability inclusion in ECCE requires systemic changes that extend beyond tool adaptation. Young children may experience a wide range of disabilities or developmental delays, each with distinct implications for learning, communication and participation in ECCE centres. Assessing these variations is complex and measuring disability inclusion in early childhood education settings in LMICs is still an underdeveloped area in international research and evaluation.²¹

²¹ Banks and Zuurmond (2015); Light for the World (2021).

4. Conclusions

Tool adaptation must be guided by both the characteristics of the measurement tools and the specific context in which they are applied. In the ECCE outcomes funds in Rwanda and Sierra Leone, the structural quality, process quality and child development tools each required a tailored approach. While some tools like BEQI and IDELA are psychometrically validated and therefore allow only limited modifications, others, such as the structural quality checklists created for each programme, offered greater flexibility for contextualization. This variation shaped the adaptation strategy and underscored the importance of iterative and contextualized processes.

Because BEQI and IDELA are psychometrically validated tools, even minor modifications can affect their reliability and validity. Their ability to accurately measure latent constructs such as ECCE quality and child development relies on extensive validation of item properties and statistical relationships. For instance, BEQI data from over 800 ECD settings in 16 countries has confirmed that three sub-scales — play-based learning, learning through conversations and promoting strong relationships — form a coherent internal structure within the broader construct of “quality.”²² Similarly, IDELA’s four-domain model — early numeracy, literacy, socio-emotional and motor skills — has been validated across five countries and shown to predict reading and mathematics outcomes.²³

In comparison, the structural quality tool had a greater scope for contextualization, as it was developed from national standards specific to each programme. However, this required balancing measurement comprehensiveness and feasibility. While checklist-like tools are useful for aligning with policy goals, some items need to be broken down into observable, low-level criteria that can be reliably assessed by enumerators with limited ECCE expertise. For example, the item: “Is the indoor area safe for young children?” would involve a complex and subjective judgement by the enumerator and would be potentially difficult to train on and to ensure that all enumerators judge this in the same way.

To address the challenges posed by varying scopes of permissible adaptations, the five-stage adaptation process emphasized contextualization and continuous stakeholder engagement. Initial consultations with tool creators and national stakeholders were central to understanding acceptable modifications and gathering iterative feedback. Pre-pilot and pilot testing helped assess usability and informed refinements for the structural quality tool, such as the restructuring of ambiguous items into clearer, more actionable indicators. These stages also generated the necessary data to test the continued reliability of the psychometrically validated tools. Finally, clear communication of the adaptations was essential to build consensus and to ensure that stakeholders understood what criteria implementers were being evaluated on.

The iterative and participatory nature of this adaptation process enabled continuous improvement and supported the achievement of four primary goals: aligning tools with national systems; maintaining psychometric soundness; ensuring usability and feasibility; and meeting the operational requirements of OBF.

²² Love et al. (2005).

²³ Halpin et al. (2019); Wolf et al. (2017).

4.1 Returning to the objectives

This section summarizes how these objectives were met and offers lessons to inform future tool adaptation efforts.

► **Objective 1: alignment with national systems**

Ensuring alignment with national systems was central to the tool adaptation process and critical to the long-term sustainability of the ECCE outcomes funds. To support institutional uptake, tools were modified to reflect ECCE policy priorities, national curricula and minimum quality standards in Rwanda and Sierra Leone. This was achieved through document reviews, expert consultations and stakeholder engagement at each stage. Participatory processes such as item review workshops and pre-testing were essential for integrating local knowledge and refining tools based on country-specific insights. While core standardized indicators were maintained to preserve psychometric validity, each tool was adapted to reflect local needs, challenges and policy frameworks. Continuous engagement with tool creators, such as ECD Measure, the creator of BEQI, helped inform whether additional items were measurable and trainable by providing lessons from BEQI adaptation processes in other contexts.

Iterative adaptations helped bridge gaps between the selected tools, national standards and stakeholder priorities. In Rwanda, for example, the child development tool was adapted to include English-language competencies that reflected national early learning goals by adding items to IDELA. Pre-tests and pilots helped validate whether the adapted tools captured the intended outcomes in real-world settings and final consultations ensured alignment with national priorities. By planning for sustained use from the outset, including securing government endorsement and reflecting national expectations, the project laid the foundation for the tools' integration into national monitoring and quality assurance systems.

► **Objective 2: maintaining psychometric soundness**

Preserving the validity, reliability and construct alignment of psychometrically validated tools, specifically BEQI and IDELA, was essential throughout the adaptation process. This was especially critical given their role in OBF, where accurate and consistent measurement directly affects payment. A rigorous adaptation process, especially during the pilot phase, is crucial to establishing the credibility of the tools and is just as important as the main evaluation itself.

High-quality enumerator training was central to ensuring reliable findings during the pilot. Scaffolded learning, mock assessments, real-time supervision and back-up staff ensured that enumerators were well-prepared to administer the tools consistently. Pilot data was used to assess internal consistency, floor/ceiling effects and inter-rater reliability, with the tools achieving strong IRR scores (90%) across both countries. However, the pilot phase was limited by small, somewhat homogenous samples, restricting advanced psychometric testing. In particular, inclusive measurement remained a challenge during piloting, as standard pilot samples rarely include sufficient numbers of children with disabilities.

► **Objective 3: ensuring the tools' usability**

Feasibility was assessed at multiple stages of the adaptation process to ensure that the tools would be practical and reliable for enumerator implementation in the specific contexts of Rwanda and Sierra Leone. From the inception phase, expert consultations helped identify the most suitable tools for measuring project metrics. Participatory processes, such as item review workshops (Stage 2) and pre-testing (Stage 3) were crucial for integrating local knowledge and refining tools based on country-specific insights. Pre-piloting specifically helped ensure the tools were as strong as possible ahead of the pilot and allowed for the identification of practical challenges and refinement

of tool content. Pilot fieldwork (Stage 4) further tested the usability of the tools under real-world conditions, uncovering logistical realities that informed necessary adjustments to both tool design and deployment strategies. For example, enumerators were able to document challenges such as travel time between sites, centre operating schedules and the specificities of conducting home visits, particularly in rural areas. Documenting how these challenges were navigated helped preparation ahead of the programmes' implementation and impact evaluations.

► **Objective 4: compatibility with OBF**

Broad-based stakeholder engagement was essential for preparing the tools for use in OBF programmes. Measurement tools in OBF programmes must be credible and tightly aligned with programme goals to support fair and transparent payments, mitigate risks of gaming and generate actionable data for implementers. To meet these requirements, the ECCE tools that were selected were the ones deemed by EOF as best-suited for use in OBF to produce objective and verifiable results. Participatory processes ensured that tools would be relevant, measurable and feasible in practice. Although navigating diverse perspectives introduced complexity, it would ultimately strengthen the legitimacy, quality and shared ownership of the final tools, which would be critical to their success in an OBF setting reliant on trust-based outcomes partnerships.

Implementer participation was especially valuable for grounding tools in practical realities, such as the conditions in local centres and the feasibility of achieving targets. However, as in any evaluation where tools are used to assess implementer performance, caution is needed to manage incentives. This is particularly important in OBF settings given the financial stakes tied to evaluation. To mitigate misaligned incentives, OBF programmes may opt not to share tools with implementers. In practice, this is not always feasible, as many tools, particularly in their original, non-adapted forms, are placed in the public domain. In this process, implementers were given access to the original tools to promote transparency, trust and alignment. To further safeguard the integrity of measurement, a small item bank with alternate versions of selected child development tool items was developed to reduce the likelihood of “teaching to the test”.

4.2 Final reflections

The five-stage adaptation of measurement tools for use in the ECCE outcomes funds in Rwanda and Sierra Leone aims to provide a practical and adaptable framework for others undertaking similar efforts. The five stages were intentionally designed to balance technical rigour with contextual relevance and to foster stakeholder ownership throughout the process. These elements ensured that the tools were aligned with national systems, reliable, usable, and compatible with OBF. Several strategies contributed to this success:

- **Early and sustained engagement with tool creators** ensured that modifications aligned with national priorities while preserving technical integrity and psychometric principles.
- **Government-endorsed stakeholder consultations**, including consultations with programme implementers, fostered buy-in, ensured alignment with national systems and supported the potential for long-term institutionalization.
- **Rigorous pre-testing and piloting**, with carefully designed enumerator training and real-time supervision, helped refine the tools to meet usability and reliability objectives.
- **Transparent final consultations** grounded in reliable pilot data enabled consensus on the final tools, supporting the trust in credible measurement needed for successful OBF programmes.

A rigorous tool adaptation process is critical to OBF programmes which require credible and transparent alignment between measurement and programme goals to ensure fair payments, foster

stakeholder learning, and drive programme success. The five-stage measurement tool adaptation process for the ECCE outcomes funds in Rwanda and Sierra Leone successfully produced tools that can not only measure the programmes' predefined outcomes while remaining grounded in national systems but are also reliable, usable and relevant to OBF. Rather than a rigid or prescriptive model, the five-stage framework offers practical, adaptable guidance for researchers and practitioners undertaking similar efforts. In addition, EOF is committed to reflective learning and continuous improvement in designing outcomes-based approaches to ECCE and to disseminating insight generated from this process. As the adapted tools are implemented to evaluate and generate learnings from the outcomes funds in Rwanda and Sierra Leone, EOF will continue to document and share insights to support ongoing dialogue and knowledge at the intersection of ECCE and OBF.

Appendices

Appendix A. Key resources (non-exhaustive)

Anderson, Kate, and Rebecca Sayre, 'Measuring Early Learning Quality and Outcomes in Tanzania: Institutional Assessment for Integrating Early Childhood Measurement in the Pre-Primary System', *ERIC*, 2016, No. ED583018, <<https://eric.ed.gov/?id=ED583018>>.

Banks, Lena Morgon and Maria Zuurmond, *Barriers and enablers to inclusion in education for children with disabilities in Malawi*, 2015, <https://www.lshtm.ac.uk/sites/default/files/2019-06/Barriers-enablers-inclusion-Malawi_0.pdf>.

Betancur, Laura, Carolina Maldonado-Carreño, Elizabeth Votruba-Drzal, and Raquel Bernal, 'Measuring preschool quality in low- and middle-income countries: Validity of the ECERS-R in Colombia', *Early Childhood Research Quarterly*, vol. 54, 2021, pp. 86–98, <<https://doi.org/10.1016/j.ecresq.2020.08.001>>.

Chen, Si, and Sharon Wolf, 'Measuring the Quality of Early Childhood Education in Low- and Middle-Income Countries', *Frontiers in Psychology*, vol. 12, 2021, <<https://doi.org/10.3389/fpsyg.2021.774740>>.

Draper, Catherine E., Aisha K. Yousafzai, Dana C. McCoy, et al. 'The Next 1000 Days: Building on Early Investments for the Health and Development of Young Children', *The Lancet* vol. 404, no. 10467, 2024, pp. 2094–2116, <[https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(24\)01389-8/abstract](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(24)01389-8/abstract)>.

Early Learning Partnership, *Measuring Child Development and Early Learning*, 2016, <<https://documents1.worldbank.org/curated/en/659701473955877219/pdf/108286-REVISED-PUBLIC-ELP-IB4-MeasuringCD-v7-CEP.pdf>>.

Education Outcomes Fund (EOF), *New Information Brief on Outcomes Funds for Education*, 2024, <<https://www.educationoutcomesfund.org/post/new-information-brief-on-outcomes-funds-for-education>>.

Education Outcomes Fund (EOF), *Outcomes-Based Finance and Early Childhood Care and Education Concept Paper*, 2025a, <<https://www.educationoutcomesfund.org/post/outcomes-based-finance-early-childhood-care-and-education>>.

Education Outcomes Fund (EOF), *Assessing and Selecting Measurement Tools for OBF Programmes for ECCE*, 2025b, <<https://www.educationoutcomesfund.org/post/new-resources-on-selecting-measurement-tools-for-obf-for-ecce>>.

Government of Sierra Leone, Ministry of Basic & Senior Secondary Education, *National Curriculum Framework & Guidelines for Basic Education*, 2020, <<https://mbsse.gov.sl/wp-content/uploads/2021/07/Basic-Education-Curriculum-Framework.pdf>>.

Government of Sierra Leone, *Sierra Leone Education Sector Plan, Transforming Learning for All, 2022 to 2026*, 2022, <<https://www.unicef.org/sierraleone/media/1306/file/Sierra%20Leone%20Education%20Sector%20Plan%202022%20-%202026.pdf>>.

Gustafsson-Wright, Emily, Sophie Gardiner, Katie Smith, Brookings, *Ensuring Effective Outcome-Based Financing in Early Childhood Development*, 2016, <<https://www.brookings.edu/articles/ensuring-effective-outcome-based-financing-in-early-childhood-development>>.

Halpin, P. F., S. Wolf, H. Yoshikawa, et al., 'Measuring early learning and development across cultures: Invariance of the IDELA across five countries', *Developmental Psychology*, vol. 55, no. 1, 2019, pp. 23–37 <<https://doi.org/10.1037/dev0000626>>.

Holler, Siegrid, Lucio Haro, Juan Camilo Villalobos, et al., *Designing Results-Based Financing Model: Recommendations and Guidelines*, 2020, <<https://www.instiglio.org/wp-content/uploads/2023/02/Designing-a-Results-Based-Financing-Model-Recommendations-and-Guidelines.pdf>>.

International Development and Early Learning Assessment, Save the Children, *Adaptation and Administration Guideline*, 2019, <https://resourcecentre.savethechildren.net/pdf/newAdaptation-Administration-Guide-and-IDELA-Tool_2019-v2.pdf>.

Kellaghan, Thomas, and Vincent Greaney, *Public Examinations Examined*, World Bank, 2020, <<https://documents1.worldbank.org/curated/en/208231572358210940/pdf/Public-Examinations-Examined.pdf>>.

Light for the World, *Leave No Child Behind: Global Report on Inclusion in Early Childhood Education*, 2021, <<https://www.light-for-the-world.org/publications/leave-no-child-behind-invest-in-the-early-years/>>.

Love, John M., Ellen Eliason Kisker, Christine Ross, et al., 'The effectiveness of early head start for 3-year-old children and their parents: lessons for policy and programs', *Developmental Psychology*, vol. 41, no. 6, 2005, pp. 885–901, <doi: 10.1037/0012-1649.41.6.88>.

Organisation for Economic Co-operation and Development (OECD), *Starting Strong IV: Monitoring Quality in Early Childhood Education and Care*, 2015, <https://www.oecd.org/content/dam/oecd/en/publications/reports/2015/10/starting-strong-iv_g1g530b3/9789264233515-en.pdf>.

Pisani, Lauren, Ivelina Borisova, Amy Jo Dowd, Save the Children, *International Development and Early Learning Assessment Technical Working Paper [Technical Report]*, 2015, <https://idela-network.org/wp-content/uploads/2017/06/IDELA-technical-working-paper_Q4-2015.pdf>.

Pisani, Lauren, Ivelina Borisova, Amy Jo Dowd, Save the Children, *International Development and Early Learning Assessment Technical Working Paper [Technical Report]*, 2017, <https://resourcecentre.savethechildren.net/pdf/idela_technical_working_paper_2017.pdf>.

Raikes, Abbie, Natalie Koziol, Dawn Davis, and Anna Burton, 'Measuring quality of pre-primary education in sub-Saharan Africa: Evaluation of the Measuring Early Learning Environments scale', *Early Childhood Research Quarterly*, vol. 53, 2020, pp. 571–585 <<https://doi.org/10.1016/j.ecresq.2020.06.001>>.

Raikes, Abbie, Marcus Waldman, Rebecca Sayre Mojgani, Jem Alvarenga Lima, *BEQI: Evidence for Subscales*, 2025, <<https://static1.squarespace.com/static/65f38c1951a7743d4bc0f56b/t/677eff6bbb189b7a28a83ad3/1736376172055/BEQI+Brief+on+CFA+and+EFA-FINAL.pdf>>.

Republic of Rwanda, National Child Development Agency, *National Standards for the Early Childhood Development in Rwanda*, 2024, <<https://www.ncda.gov.rw/index.php?elD=dumpFile&t=f&f=137325&token=8606308b4cc24e43c085f82c29b39fa7c042f35f>>.

Shavitt, Ilana, Monica Ayres de Araujo Scatollin, Adriana Suzart Ungaretti Rossi, et al. 'Transcultural adaptation and psychometric properties of the International Development and Early Learning Assessment (IDELA) in Brazilian pre-school children', *International Journal of Educational Research Open*, vol. 3, 2022, no. 100138 <<https://doi.org/10.1016/j.ijedro.2022.100138>>.

Thorne, Georgia, Lindsay Adams, Arushi Terway, NORRAG, *Practice Note: Measuring Outcomes in Outcomes-Based Financing for Early Childhood Care and Education*, 2024, <<https://golab.bsg.ox.ac.uk/knowledge-bank/resource-library/Measuring-Outcomes-in-Outcomes-Based-Financing-for-Early-Childhood-Care-and-Education/>>.

United Nations Educational, Scientific and Cultural Organization (UNESCO), *The Right to a Strong Foundation. Global Report on Early Childhood Care and Education*, 2024, <<https://www.unesco.org/en/articles/right-strong-foundation-global-report-early-childhood-care-and-education>>.

Winthrop, Rebecca, and Eileen McGivney, *Skills for a Changing World: Advanced Quality Learning for Vibrant Societies*, 2016, <https://www.brookings.edu/wp-content/uploads/2016/05/global_20160809_skills_for_a_changing_world.pdf>.

Wolf, Sharon, Peter Halpin, Hirokazu Yoshikawa, et al. *Assessing the Construct Validity of Save the Children's International Development and Early Learning Assessment (IDELA)*, 2016, <https://resourcecentre.savethechildren.net/document/assessing-construct-validity-save-childrens-international-development-and-early-learning>.

Wolf, Sharon, Peter Halpin, Hirokazu Yoshikawa, et al., 'Measuring school readiness globally: Assessing the construct validity and measurement invariance of the International Development and Early Learning Assessment (IDELA) in Ethiopia', *Early Childhood Research Quarterly*, Vol. 41, 4th Quarter, 2017, pp. 21–36.

Appendix B. Glossary

Term	Definition
Child development	These are the developmental domains that reflect children's overall growth and readiness for school. ECCE examples include: ▶ cognition and executive function; tasks that assess memory, attention and problem-solving (e.g. sorting shapes or following multi-step instructions) ▶ language and communication; ability to understand and use language, such as naming objects or retelling a story ▶ motor skills; activities like hopping, drawing shapes or stacking blocks to assess fine and gross motor development ▶ socio-emotional development; recognizing emotions, cooperating with peers and showing empathy ▶ early literacy and numeracy; identifying letters and numbers, recognizing patterns, or counting objects.
Construct validity	The extent to which the tool measures what it is intended to measure (i.e. the items meaningfully capture the latent constructs). Methods such as Exploratory Factor Analysis (EFA) and Confirmatory Factor Analysis (CFA) can be used to study how well the hypothesized theoretical structure of the tool is upheld by real-world data. This allows us to make inferences such as whether the items used to study a domain are suitable or not.²⁶ It is a helpful measure to study when adapting a tool for a new context.
Domains and sub-domains	A domain is a broad organizing category that contains related latent constructs or skills within it. A sub-domain is a more specific category within a broader domain. Child development, process quality and structural quality are top-level domains. Child development includes sub-domains such as social-emotional development, which can be further broken down into lower-level domains or specific constructs such as self-awareness.²⁵
Floor and ceiling effects	Floor and ceiling effects refer to the phenomenon in which a large proportion of respondents score at the lowest (floor) or highest (ceiling) ends of a measurement scale. These effects suggest that the instrument may be too difficult, or too easy, respectively, and therefore may not adequately capture the full range of variability within the target population. Domains that are common to most, if not all children, such as use of a particular language, may result in a ceiling effect.
Internal consistency	Internal consistency is a study of correlation between items to check that the set of items intended to measure the same underlying construct yielded consistent results. To assess internal consistency and reliability, statistics such as Cronbach's alpha can be used to measure the coherence of responses within a tool.
Inter-rater reliability (IRR)	Inter-rater reliability measures the level of agreement or consistency between two or more enumerators when assessing or scoring the same event or participant, ensuring the results are not dependent on who conducts the evaluation. Inter-rater reliability is especially important for observational and assessment tools that involve classifying human behaviour. Training and documentation should guide enumerators so that they do not need to rely on subjective judgments. Inter-rater reliability measures the extent to which that guidance has been successful.

Term	Definition
Latent constructs	Underlying skills, qualities or behaviours that cannot be observed directly, so must be inferred through other observable tasks or behaviours. ECCE examples include: ▶ emotional awareness ▶ understanding of one-to-one correspondence ▶ print awareness ▶ quality of child-caregiver interactions.
Outcomes-based financing (OBF)	Outcomes-based financing is a sub-category of results-based financing (RBF) where the pre-agreed results focus primarily on outcomes, including intermediate, short-term and medium-term final outcomes for programme participants.
Outcomes funds	Outcomes funds are a form of outcomes-based financing, where funding is pooled to make outcomes-linked payments to multiple programme implementers working towards the same, predefined outcomes. Funds are released upon completion of agreed upon outcomes.
Pilot	Pilot testing a measurement tool is the process of trying out a newly adapted tool on a small sample of the target population before full implementation, to check whether the items are understandable, culturally appropriate, feasible to administer and able to capture the intended information. The pilots are carried out across a large number of centres, during which time any adaptations are made and included in future programmes.
Pre-test	The pre-testing stage serves as a bridge between the theoretical design of the tools and their practical application within the real-world settings of ECCE centres. In this brief, pre-testing refers to the initial test of the tools in a limited number of centres, whereas the pilot represents a full test of the tools in a larger number of centres during which data is gathered for analysis. The pre-tests are carried out on a small number of centres initially, during which time any adaptations are made and included in the pilot.
Process quality	Educator practices and their interactions with learners. ECCE examples include: ▶ educator-child interactions; teachers engaging children in conversation, asking open-ended questions and encouraging self-expression ▶ learning activities; use of play-based learning, storytelling and group activities that promote cognitive and social development ▶ emotional support; educators showing warmth, patience and responsiveness to children's needs and emotions.
Psychometrics	The theory and methods used to design measurement tools in psychology. Psychometrically validated tools measure what they are intended to measure and produce consistent results over space and time. This enables confidence in their accuracy, supports tracking change over time and allows for comparisons across settings, such as countries or programmes.

Term	Definition
Scales or tasks	Scales or tasks are groupings of one or more items (single responses) which together allow us to draw broader inferences about domains, sub-domains or specific constructs. In IDELA, the task on shape identification consists of multiple items that check for the child's ability to identify individual shapes. In the process quality tool, the sub-scale on learning through conversation consists of individual items that capture opportunities for dialogue, self-expression and comprehension.
Structural quality	The regulable resources in learning settings. ECCE examples include: ▶ physical environment; availability of safe and clean classrooms, age-appropriate furniture and outdoor play areas ▶ educator characteristics; educators with formal ECCE training or certification ▶ child-to-educator ratio; maintaining a recommended number of children per educator.

Appendix C. Sample adaptations in structural quality tool in Sierra Leone

Domain	Item (EOF original checklist)	Change in item post-pilot	Updated item
Basic sanitation	Do children wash their hands after toileting?	Dropped. Moved to process quality tool.	N/A.
Basic sanitation	Are there separate toilet facilities for boys and girls?	Rephrased.	Are toilet facilities for children separated by gender?
Class size/staff characteristics	Number of centre teachers with an ECCE certification of qualification >0	Rephrased.	At least one educator has ECD training.
Safety/hazard-free	Is the building clean and safe for young children?	Broken down into multiple items.	(a) Is the outdoor space clean? (b) No dangerous objects in outdoor space. (c) No sharp objects sticking out of centre walls. (d) No broken walls, doors or windows outdoors.
Safety/hazard-free	Is there an outdoor playground/outdoor play area that is clean?	Rephrased.	There is a safe outdoor play area.
Safety/hazard-free	Within the fenced-in area, is the land dry and not waterlogged?	During the item review workshop, it was suggested to delete this item. With five to six months of rain in SL, it is difficult to find a centre area that is dry. Also, waterlogging is not an issue in rural areas. If there is a flood, the centre is going to be closed for observation anyway and there is little that the ECD centre can do about it.	N/A.
Space and materials	Is there an outdoor area with adequate space for play?	Rephrased.	Does the outdoor area have enough space for children to run freely?
Space and materials	Is the indoor learning space enough for all attending children to engage in all indoor activities?	Rephrased.	Is the indoor learning space enough for all attending children to play freely?
Space and materials		Added a new item.	Are there toys or materials to play with?

Appendix D. Overview and key lessons

Adapting ECCE Measurement Tools for Outcomes-Based Financing: A Five-Stage Process

Purpose: Adapt measurement tools to ensure they are relevant, robust, usable and appropriate for use in OBF

What is being measured in the outcomes funds in Rwanda and Sierra Leone:



Structural Quality: Physical characteristics related to safety, sanitation, and materials (checklist based on National Minimum Standards)



Process Quality: Educator-child interactions (BEQI)



Child Development: Assessment to measure child outcomes across five key domains (IDELA)

Adaptation process:

Document review:
Align with National Standards

Stakeholder consultations:
Gather feedback

Pre-testing:
Identify practical and operational issues

Piloting:
Tool and fieldwork model validation

Finalization and consensus:
Incorporate feedback and official sign-offs

Key lessons from adaptation process:

Include the tools' creators in the process

- Helps preserve the original intent, structure, and psychometric integrity of the tool during adaptation.
- Ensures that any changes (additions, removals, or rewording) are aligned with what the tool was designed to measure.

Consult diverse stakeholders with the government in the lead

- Builds buy-in and legitimacy by involving those who will approve and use the tools, particularly national governments.
- Incorporates knowledge from implementers and practitioners to ensure contextual relevance and usability.

Pre-test before the pilot

- Allows early identification of major usability or contextual issues before full-scale piloting makes changes costly or infeasible.
- Enables refinements to definitions, instructions, or scoring methods to improve clarity and reduce enumerator confusion.

Treat the pilot like real fieldwork

- Conducting reliability training and deploying full data management protocols ensures pilot results are valid and usable.
- Reveals how well the tool performs under realistic field conditions, allowing adjustments before finalization.

Build consensus

- Helps secure final sign-off from both technical and managerial stakeholders by demonstrating how feedback was incorporated.
- Strengthens confidence in the tools during implementation and verification (particularly important in OBF settings) and long term.

About EOF

The Education Outcomes Fund (EOF) is an ambitious effort to significantly improve learning and employment outcomes by tying funding to measurable results. Through a partnership model, we bring together governments, donors, implementing partners, and investors to achieve concrete targets for learning, skills development, and employment. Using our global platform, we are scaling up outcomes-based financing in education, with the aim of improving the effectiveness of spending and transforming the lives of 10 million children and youth.

EOF is an independent trust fund hosted by the United Nations Children's Fund (UNICEF). The Fund works to help achieve the United Nations (UN) Sustainable Development Goal 4 (SDG 4) of inclusive and equitable quality education for all. EOF was founded in 2018 by The Education Commission and GSG Impact.

